



VMWARE VSPHERE 7.0: INSTALL, CONFIGURE, MANAGE



Lab 08: Configuring Active Directory Domain Services on VMware vSphere

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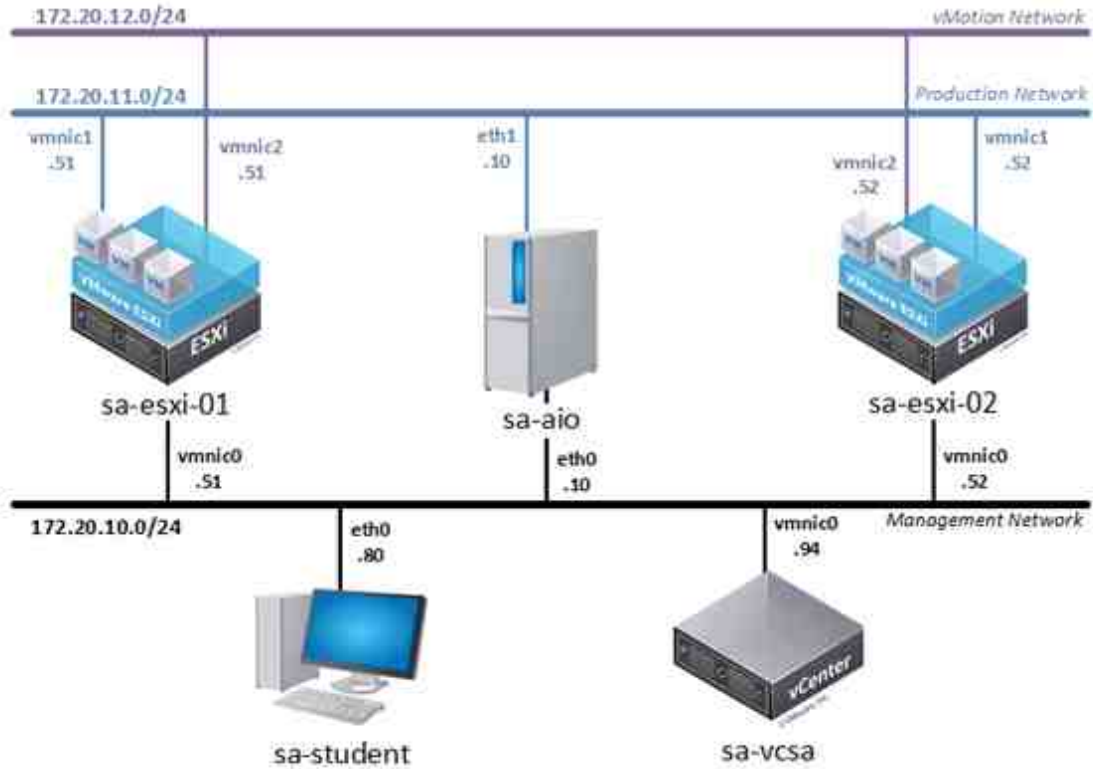
Introduction

In this lab, you will join vCenter Server to the netlab.local domain and add netlab.local as an identity source. Assign roles and permissions so that an Active Directory user can perform functions in vCenter Server.

Objectives

-) Join vCenter Server to the netlab.local domain
-) Add netlab.local as an Identity Source
-) View Active Directory users
-) Add an ESXi host to Active Directory and Assign object permission to an Active Directory user
-) Assign Root-level global permission to an Active Directory user
-) Log in as an Active Directory user
-) Use an Active Directory user to navigate the vSphere client and create a virtual machine
-) Modify permissions for an Active Directory user and navigate the vSphere Client after changing the permissions

Lab Topology



Lab Settings

The information in the table below will be needed in order to complete the lab. The task sections below provide details on the use of this information.

Virtual Machine	IP Address	Account	Password
sa-esxi-01	eth0: 172.20.10.51	root	Train1ng\$
sa-esxi-02	eth0: 172.20.10.52	root	Train1ng\$
sa-aio	eth0: 172.20.10.10	sysadmin	vmware123
sa-student	eth0: 172.20.10.80	sysadmin	vmware123
sa-vcasa	eth0: 172.20.10.94	sysadmin@vclass.local	vmware123

1 Join vCenter Server to the Netlab.local Domain

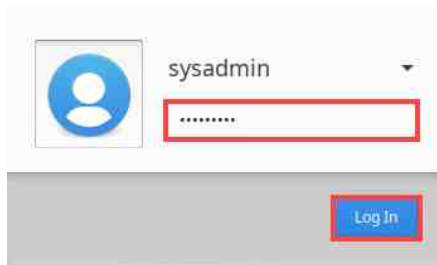
In this task, you will join 172.20.10.94 (sa-vcsa-01) to the netlab.local domain, which is an Active Directory (AD) identity source. After joining the domain, an AD user can be selected and assigned rights to manage a virtual machine.

1. Launch the sa-student virtual machine to access the graphical login screen.

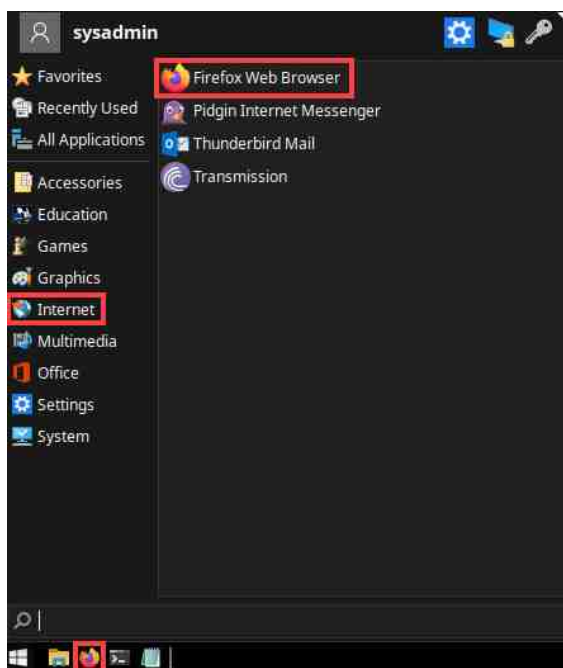


To launch the console window for a virtual machine, you may access by either clicking on the machine's graphic image from the topology page or by clicking on the machine's respective tab from the navigation bar.

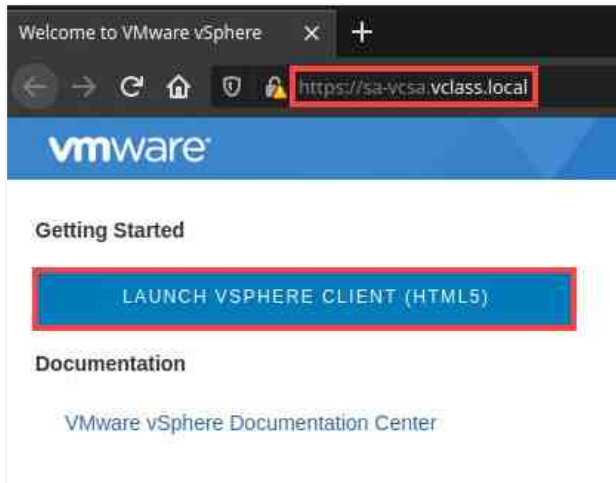
2. Log in as sysadmin using the password vmware123.



3. Launch the Mozilla Firefox web browser by either clicking on the icon shortcut found on the bottom toolbar or by navigating to Start Menu > Internet > Firefox Web Browser.

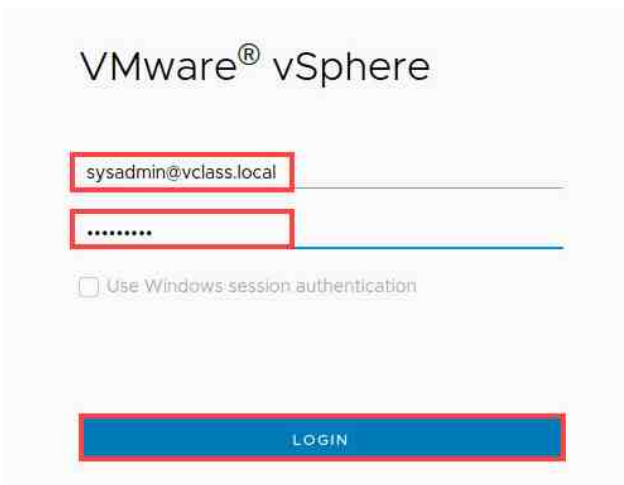


4. Notice the homepage is automatically configured to load the URL address of the VMware vCenter Server Appliance (sa-vcasa virtual machine). Click on the LAUNCH VSPHERE CLIENT (HTML5) button to launch the vSphere Client.



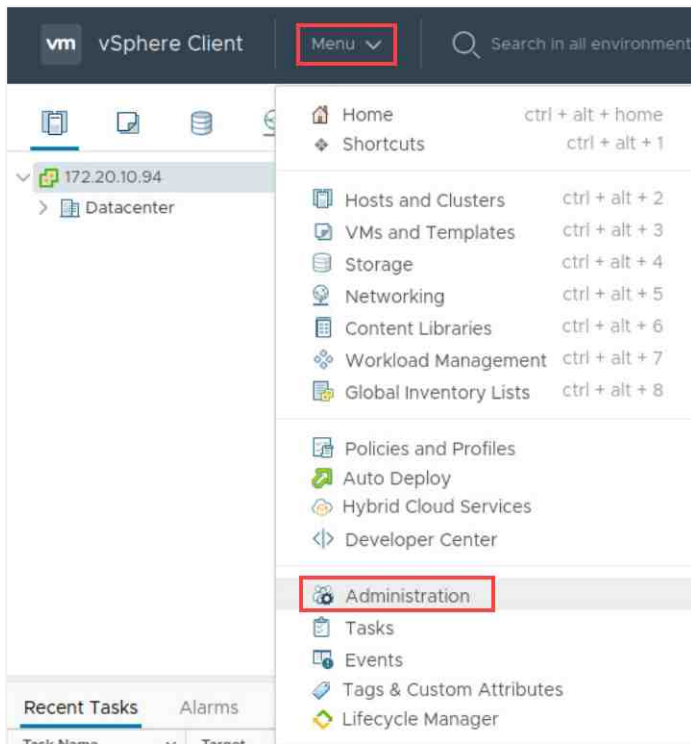
If you cannot successfully load the VMware Getting Started webpage, then please wait an additional 3-5 minutes and refresh the page to continue. During this time, the vCenter Server Appliance is still booting up and requires extra time to initialize.

5. In the Username text field, type `sysadmin@vclass.local` and in the Password text field, type `vmware123`. Click on LOGIN.



You may ignore the “browser-OS combination” warning message presented on the VMware vCenter Single Sign-On page and continue moving forward with the lab.

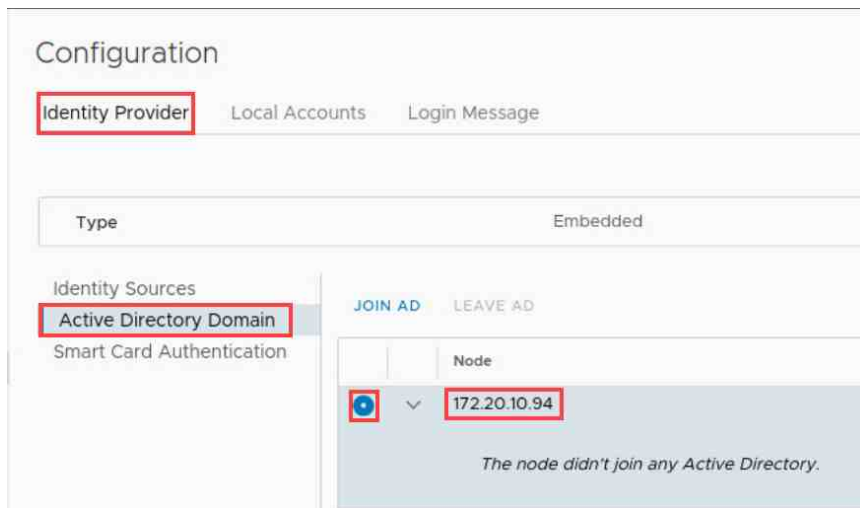
6. In the vSphere Client, from the dropdown Menu, select Administration.



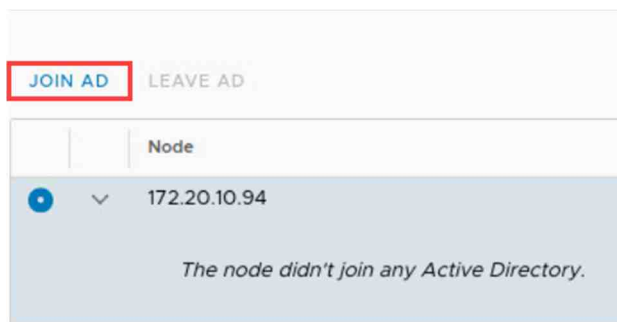
7. In the navigation pane under Single Sign On, select Configuration.



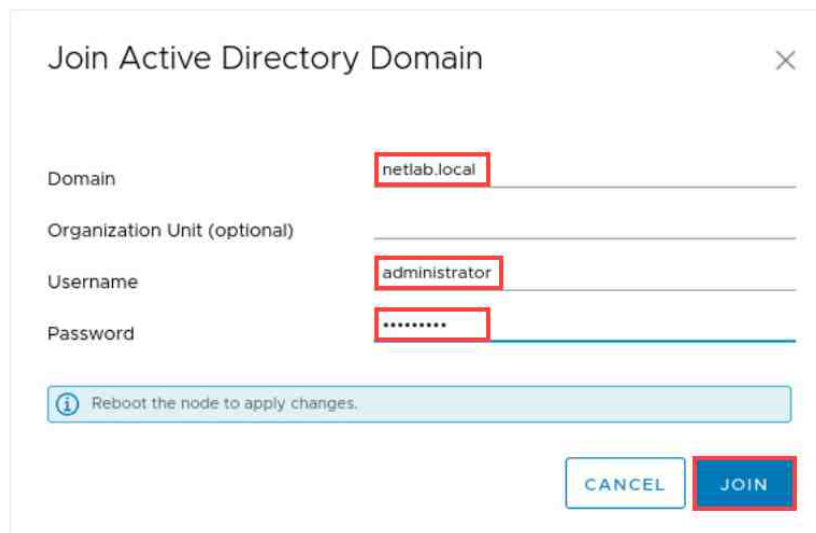
8. In the Configuration pane, select Identity Provider and click Active Directory Domain. Verify the 172.20.10.94 node is selected.



9. Click Join AD.



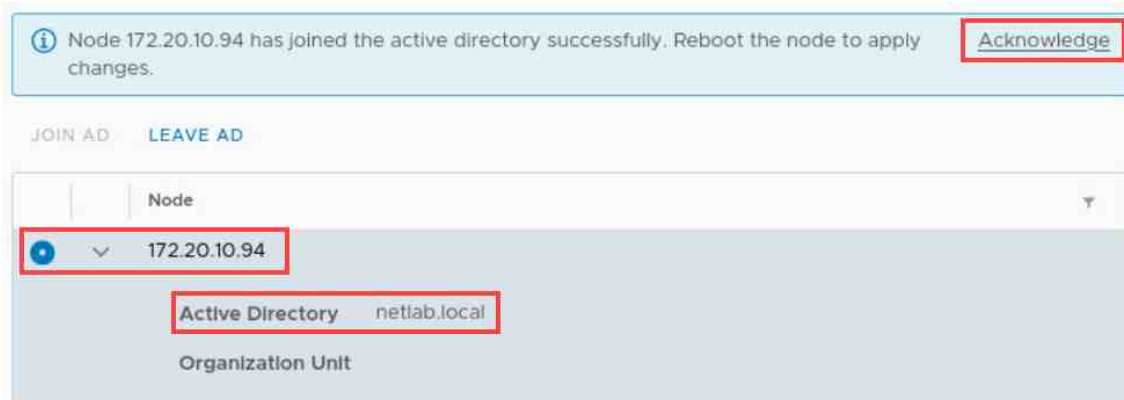
10. In the Join Active Directory Domain window, enter `netlab.local` for the Domain, `administrator` for the Username, and `Training$` for the Password. Click Join.



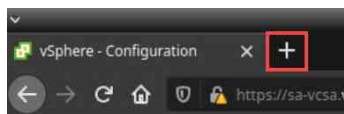
Please Note

For this lab, Active Directory has been preconfigured on the SA-AIO machine.

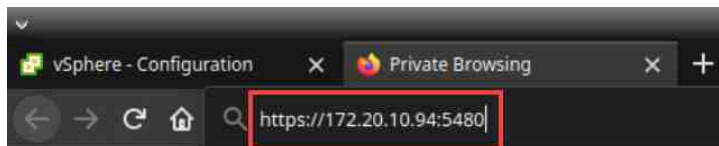
11. Verify that 172.20.10.94 has successfully joined the netlab.local Active Directory. Click Acknowledge on the popup dialog box.



12. Restart the vCenter Server Appliance using the vCenter Server Appliance Management Interface.
- a. Open a new tab in the Firefox Web Browser.

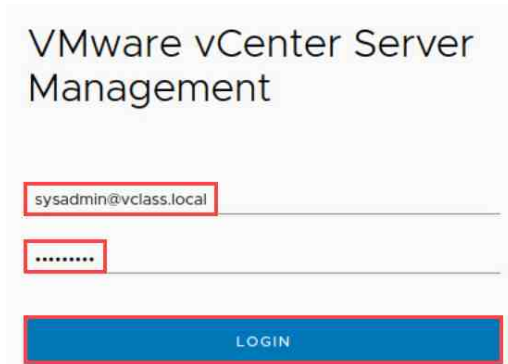


- b. Type `https://172.20.10.94:5480` in the address bar. Press Enter.



Port 5480 is the default port used to access the vCenter Server Appliance Web User Interface. The VMware vCenter Server Appliance Management Interface (VAMI) is used to perform administrative tasks such as changing the host name, network configurations, applying

- c. In the Username text field, type `sysadmin@vclass.local` and in the Password text field, type `vmware123`. Click on LOGIN.

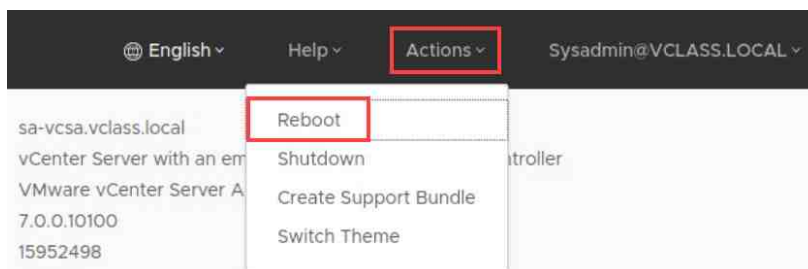


VMware vCenter Server Management

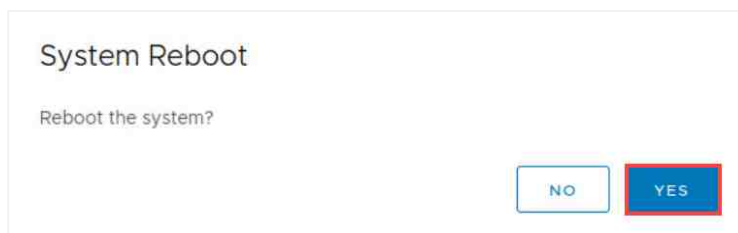
sysadmin@vclass.local

LOGIN

- d. From the Actions dropdown menu in the top-right corner, select Reboot.



- e. In the System Reboot window, click YES.



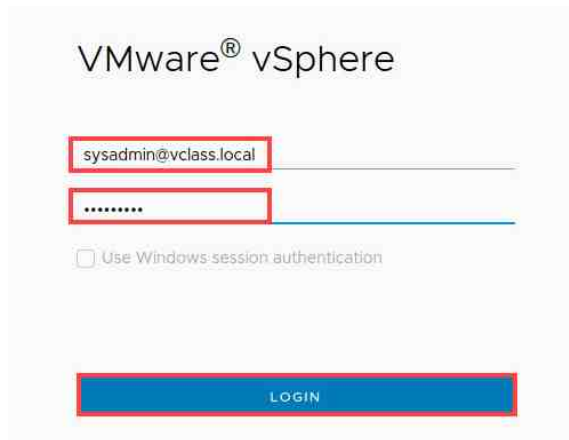
The reboot process takes 5 to 10 minutes to complete. During this time, the vSphere Client is unavailable. You will not be able to complete the next task until vCSA has fully rebooted.

13. Change focus back to the vSphere Web Client tab, refresh the screen periodically until the vSphere Client login page appears. Once the login page appears, you may continue to the next task.

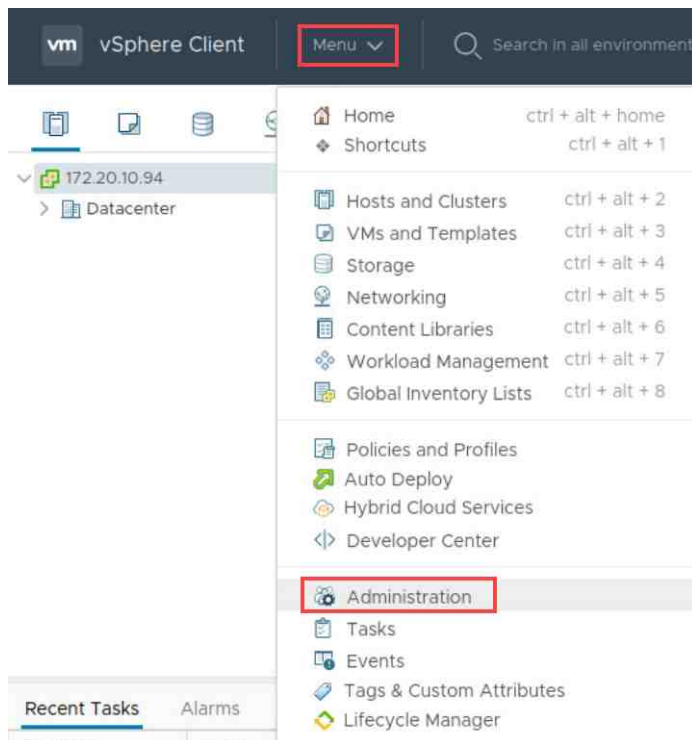
2 Add Netlab.local as an Identity Source

In this task, you will add an identity source to enable the single sign-on configuration.

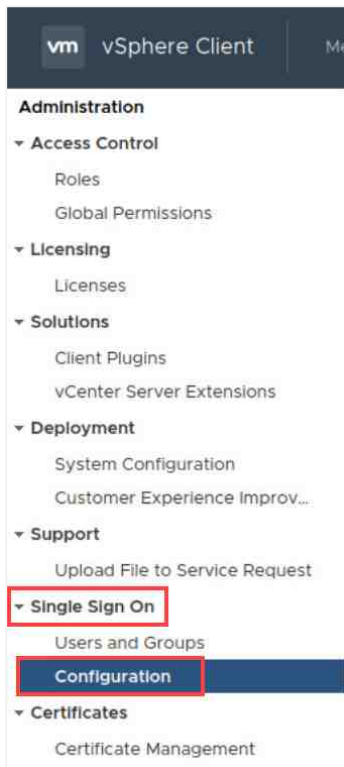
1. Log in to the vSphere Client, entering `sysadmin@vclass.local` for the Username and `vmware123` for the password.



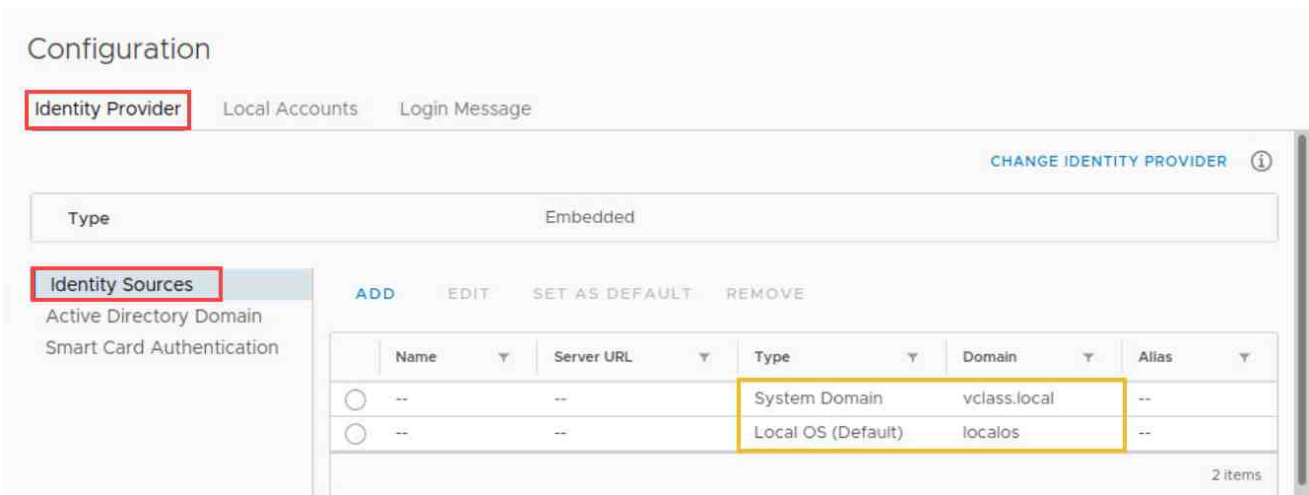
2. From the dropdown Menu, click Administration.



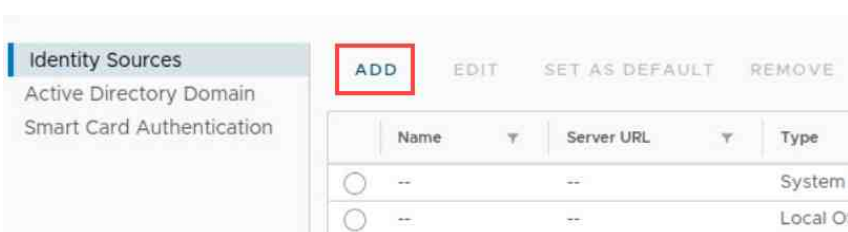
3. In the navigation pane under Single Sign On, select Configuration.



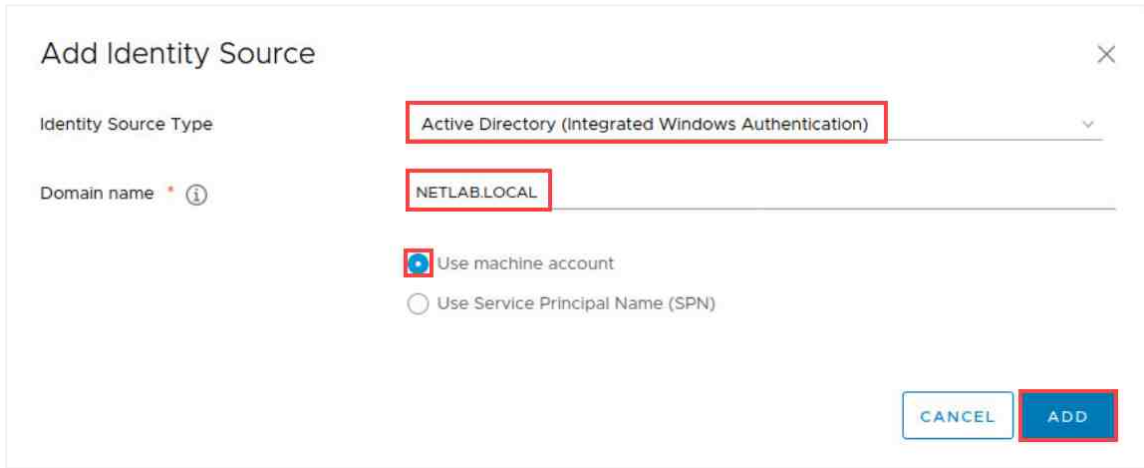
4. In the Configuration pane, select Identity Provider and click Identity Sources. Notice the vclass.local and local OS domains appear as identity sources.



5. Click ADD on the Identity Sources dialog box.



- In the Add Identity Source window, verify that Active Directory (Integrated Windows Authentication) is selected. Verify that NETLAB.LOCAL is listed as the Domain name. Ensure that Use machine account is selected and click ADD.



Add Identity Source

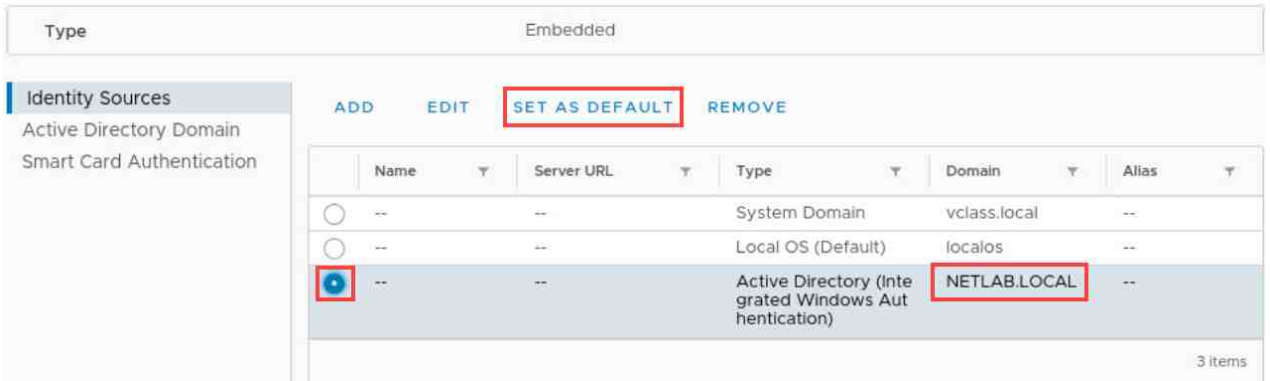
Identity Source Type: **Active Directory (Integrated Windows Authentication)**

Domain name: **NETLAB.LOCAL**

☒ Use machine account
☐ Use Service Principal Name (SPN)

CANCEL **ADD**

- In the Identity Source window, verify that NETLAB.LOCAL is listed as an identity source. Select the NETLAB.LOCAL identity source and click SET AS DEFAULT.



Type: Embedded

Identity Sources: Active Directory Domain, Smart Card Authentication

ADD EDIT **SET AS DEFAULT** REMOVE

	Name	Server URL	Type	Domain	Alias
☐	--	--	System Domain	vclass.local	--
☐	--	--	Local OS (Default)	localos	--
☒	--	--	Active Directory (Integrated Windows Authentication)	NETLAB.LOCAL	--

3 items

- In the Set Default Identity Source window, click OK.



Set Default Identity Source

 This will alter your current default domain. Do you want to proceed?

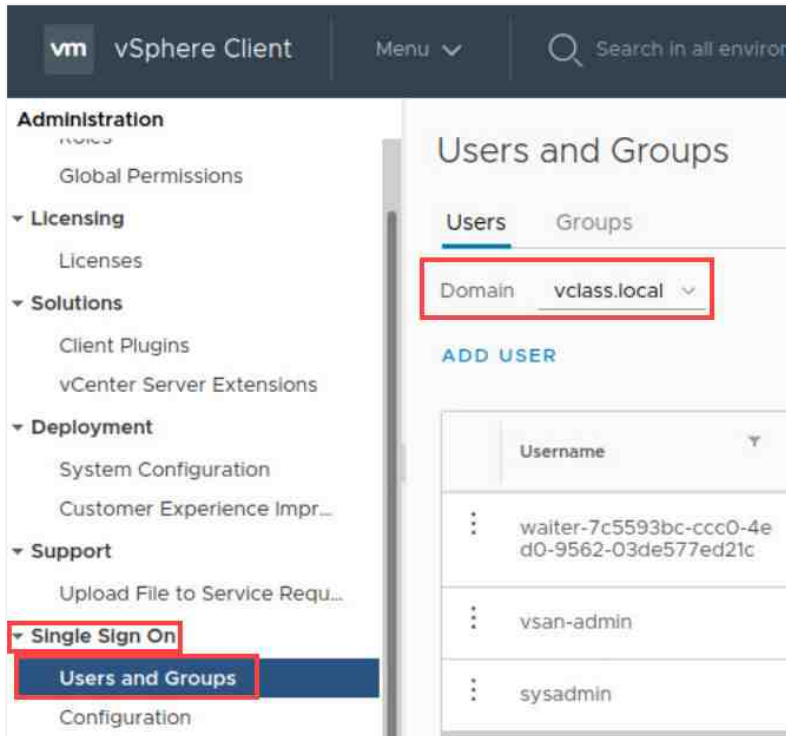
CANCEL **OK**

- Leave the vSphere Client open and continue to the next task.

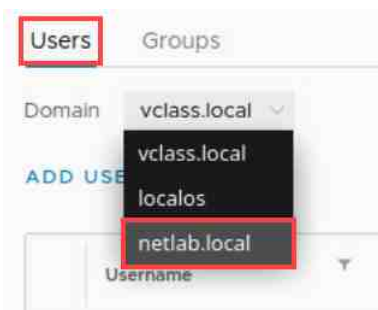
3 View Active Directory Users

In this task, you view the list of Active Directory users to verify that the Administrator and sysadmin single sign-on account exists.

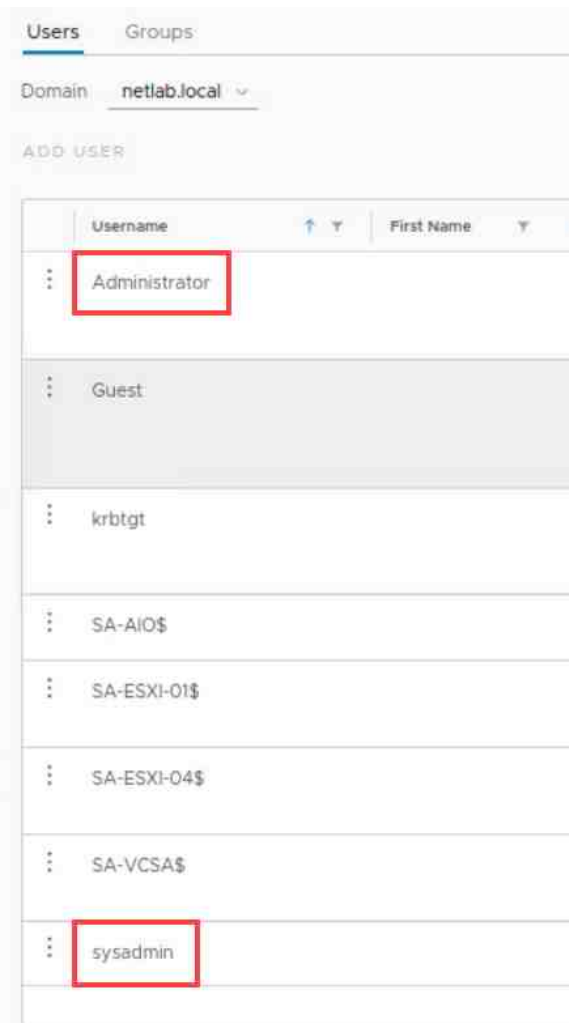
1. Under Single Sign-On, click Users and Groups. By default, the list of users for the vclass.local domain will appear in the right pane.



2. In the Users pane, select netlab.local from the Domain dropdown menu.



3. Verify that sysadmin and Administrator appear on the list.

**Please
Note**

You may need to scroll through the Users window to verify that both sysadmin and Administrator are listed.

4. Leave the vSphere Client open and continue to the next task.

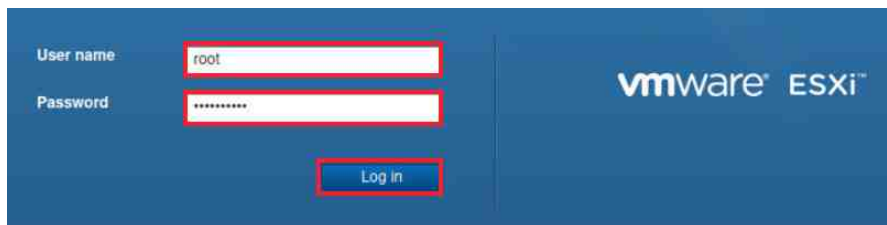
4 Add an ESXi Host to Active Directory and Assign Object Permission to an Active Directory User

In this task, you will add an ESXi host to Active Directory and assign permission at the vCenter Server level to the sysadmin@netlab.local user. This permission propagates to the child objects of vCenter Server.

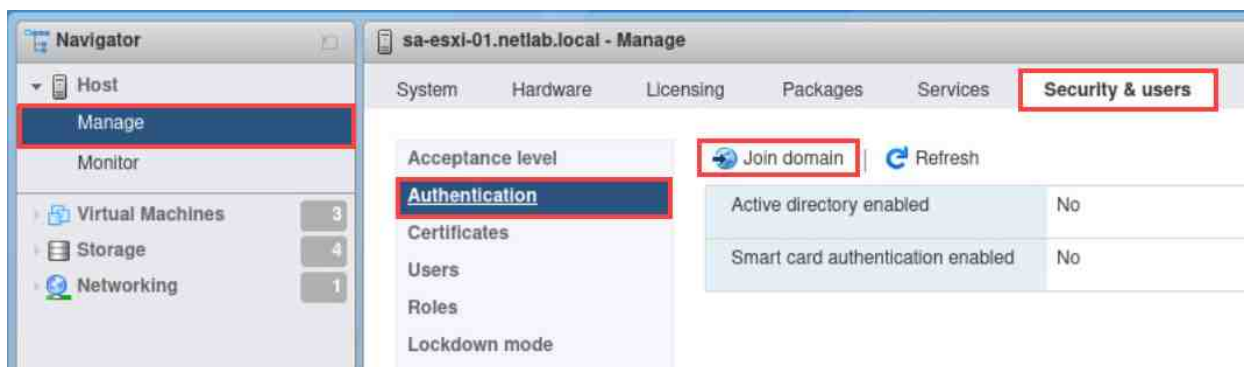
1. Open a new tab in the Mozilla Firefox browser, then type in `https://172.20.10.51` into the address field and press Enter.



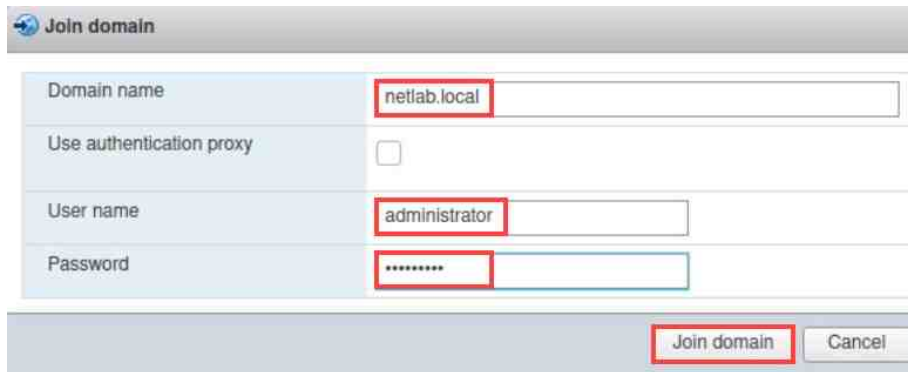
2. On the VMWARE ESXi Web Client page, type `root` as the User name and `Training$` as the Password. Click Log in.



3. In the navigator pane, click on the Manage tab. In the sa-esxi-01.netlab.local – Manage window, select the Security & users tab. Click Authentication and Join domain.

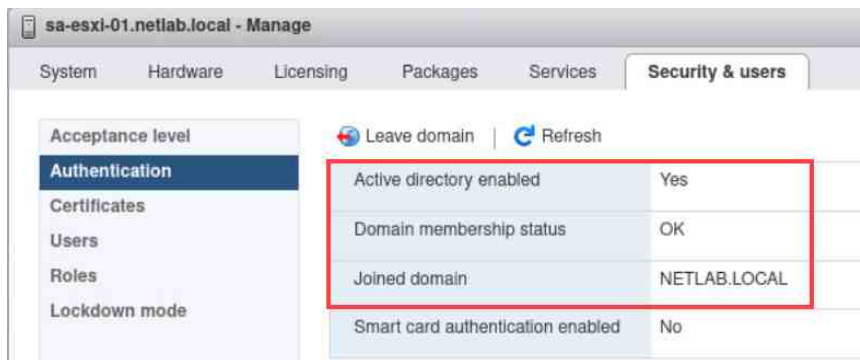


4. In the Join Domain window, type `netlab.local` for the Domain name, `administrator` for the User name, and `Training$` for the Password. Click Join Domain.



The 'Join domain' dialog box is shown. It has four input fields: 'Domain name' (containing 'netlab.local'), 'Use authentication proxy' (unchecked), 'User name' (containing 'administrator'), and 'Password' (containing '*****'). At the bottom right are 'Join domain' and 'Cancel' buttons.

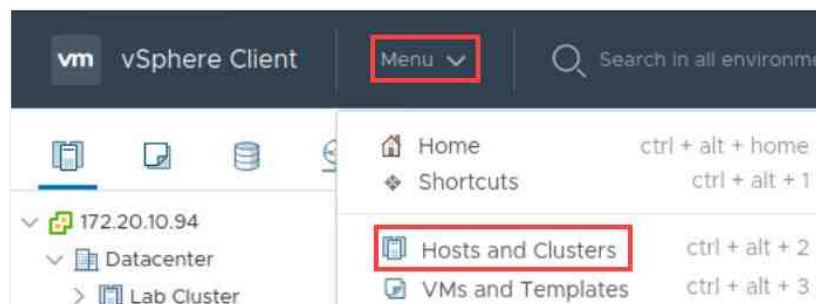
5. Verify that sa-esxi-01 has successfully joined Active Directory.



The 'sa-esxi-01.netlab.local - Manage' window is shown, with the 'Security & users' tab selected. The 'Authentication' sub-tab is active. A table shows the following status:

Active directory enabled	Yes
Domain membership status	OK
Joined domain	NETLAB.LOCAL
Smart card authentication enabled	No

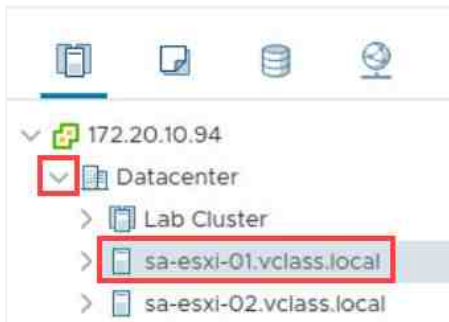
6. Change focus back to the vSphere Client. From the dropdown Menu, select Hosts and Clusters.



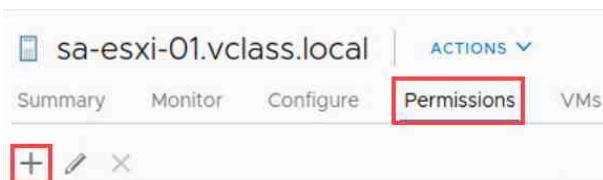
The vSphere Client interface is shown. The 'Menu' dropdown is open, and 'Hosts and Clusters' is selected. The menu options are:

- Home (ctrl + alt + home)
- Shortcuts (ctrl + alt + 1)
- Hosts and Clusters (ctrl + alt + 2)
- VMs and Templates (ctrl + alt + 3)

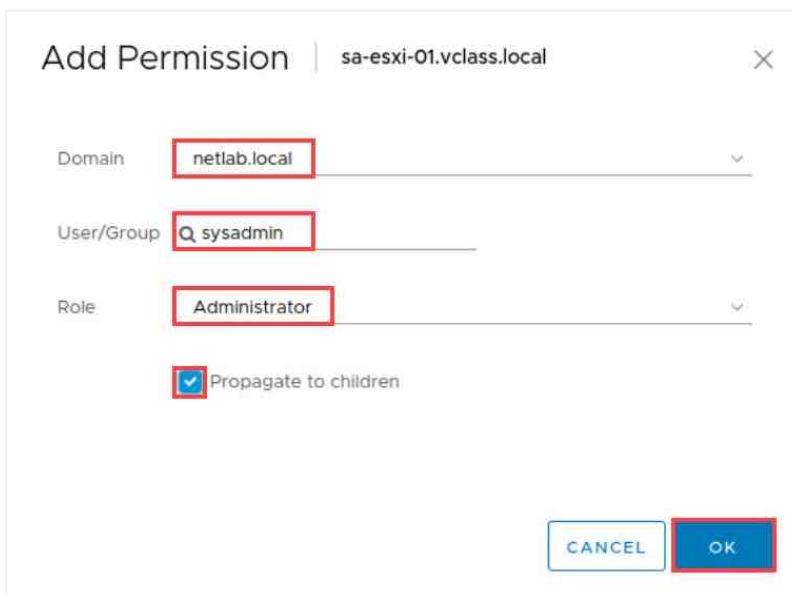
7. In the navigation pane, expand Datacenter and select sa-esxi-01.vclass.local.



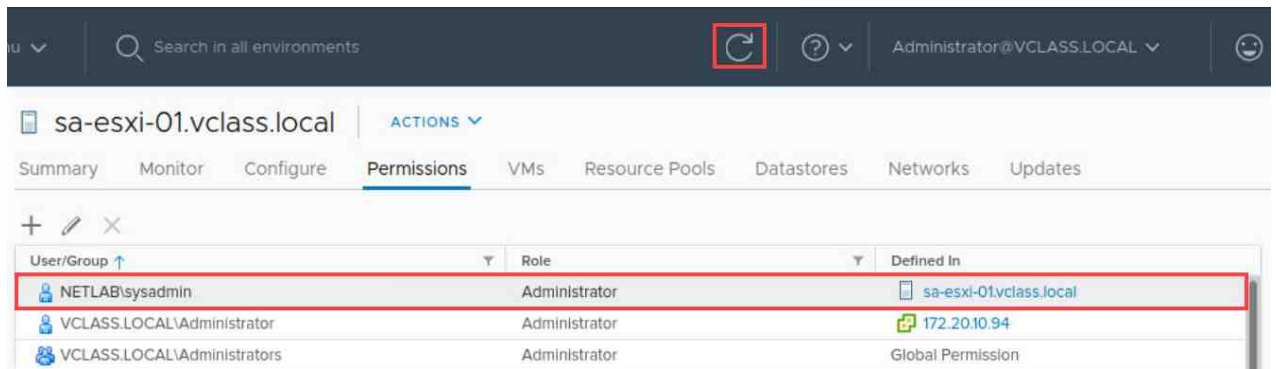
8. In the right pane, select Permissions and click Add Permission (the plus icon).



9. In the Add Permission window, from the dropdown Domain menu, select netlab.local. Type sysadmin for the User/Group and leave Administrator for the Role. Select Propagate to children and Click OK.



10. Verify that NETLAB\sysadmin appears in the list, is assigned the Administrator role, and is defined in sa-esxi-01.vclass.local, which is the vCenter Server object and its children. If the permission has not been updated in the list, click the refresh icon located at the top of the vSphere Client window.

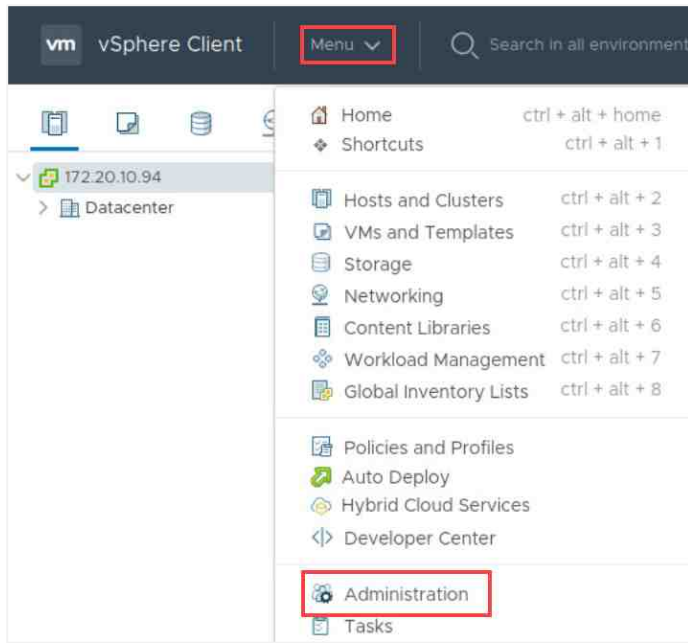


11. Leave the vSphere Client open and continue to the next task.

5 Assign Root-Level Global Permission to an Active Directory User

In this task, you will grant global permission to sysadmin@netlab.local to administer vCenter with full administrative rights.

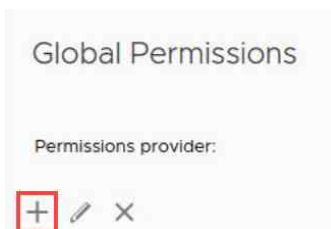
1. From the dropdown Menu, select Administration.



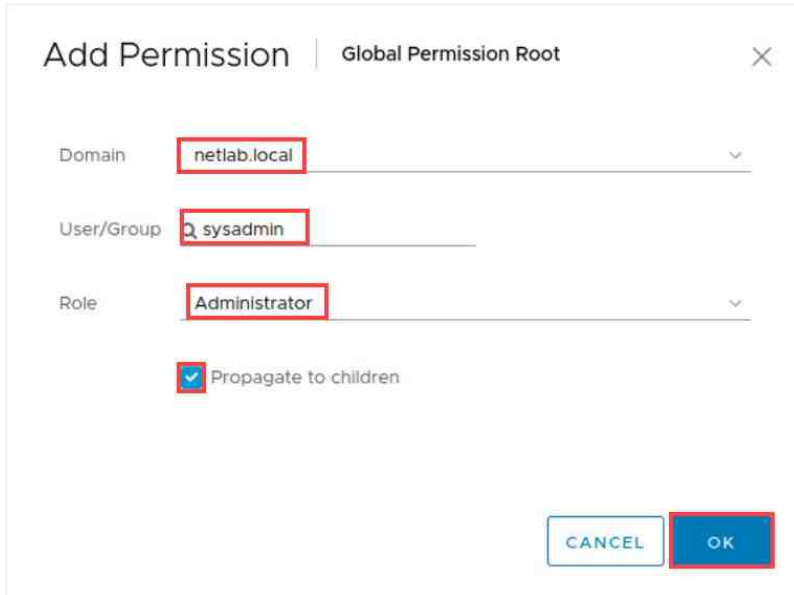
2. In the navigation pane, expand Access control and click Global Permissions.



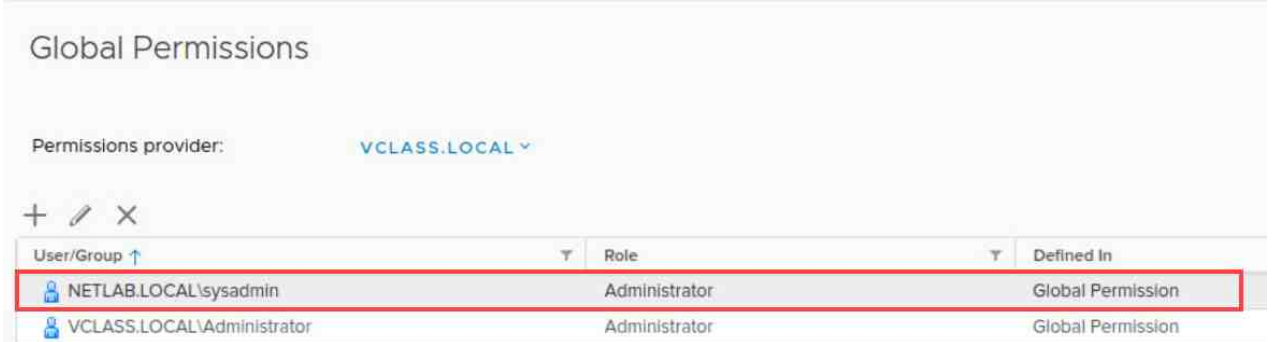
3. In the Global Permissions pane, click the Add Permission icon (the plus sign).



4. In the Add Permission window, from the dropdown Domain menu, select netlab.local. Type sysadmin for the User/Group. From the Role dropdown menu, select Administrator. Select Propagate to children and Click OK.



5. Verify that NETLAB.LOCAL\sysadmin appears in the list, is assigned the Administrator role, and is defined in the vCenter Server object as Global Permission.



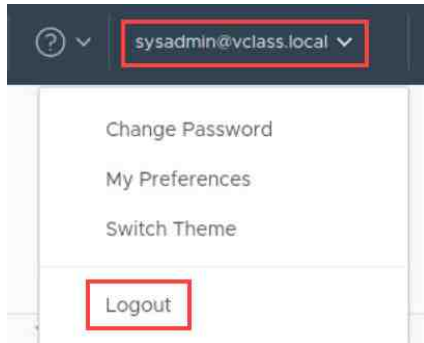
Global Permissions		
Permissions provider: VCLASS.LOCAL		
+ ✎ ✕		
User/Group ↑	Role	Defined In
NETLAB.LOCAL\sysadmin	Administrator	Global Permission
VCLASS.LOCAL\Administrator	Administrator	Global Permission

6. Leave the vSphere Client open and continue to the next task.

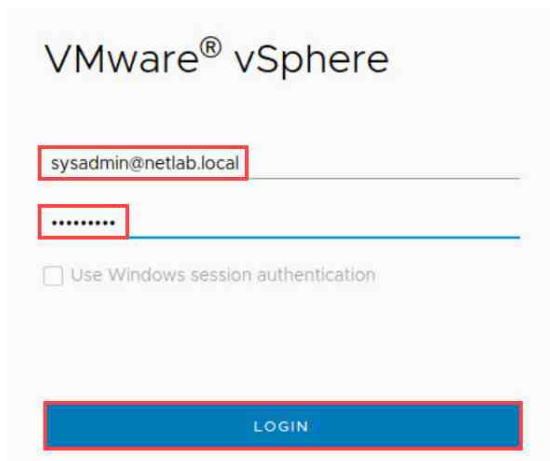
6 Log in as an Active Directory User

In this task, you will log in as `sysadmin@netlab.local` and verify that the login is successful.

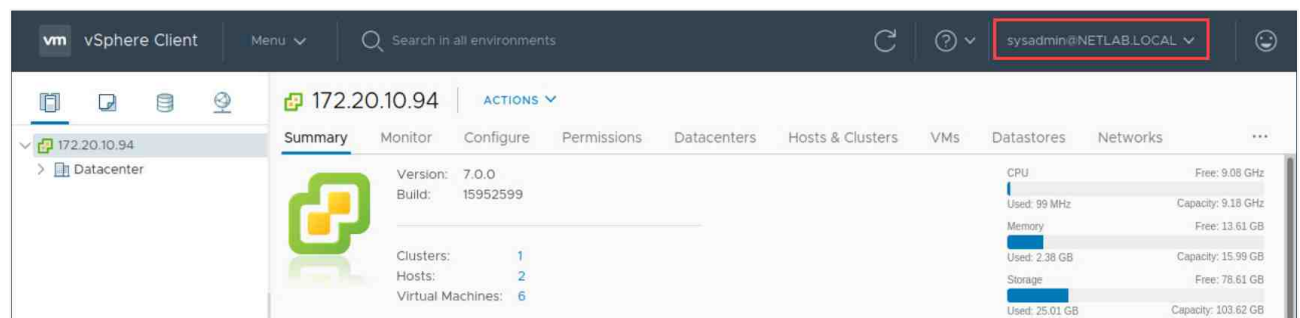
1. Log out of the vSphere Client by selecting `sysadmin@vclass.local` and click Logout.



2. On the vSphere Client login screen, enter `sysadmin@netlab.local` and `vmware123` as the password. Click LOGIN.



3. Verify that you are logged in to the vSphere Client as `sysadmin@netlab.local`.

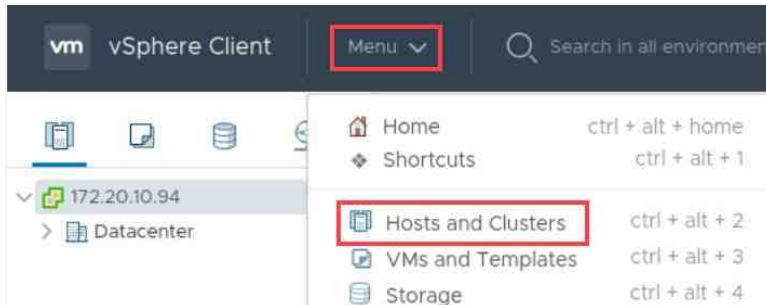


4. Leave the vSphere Client open and continue to the next task.

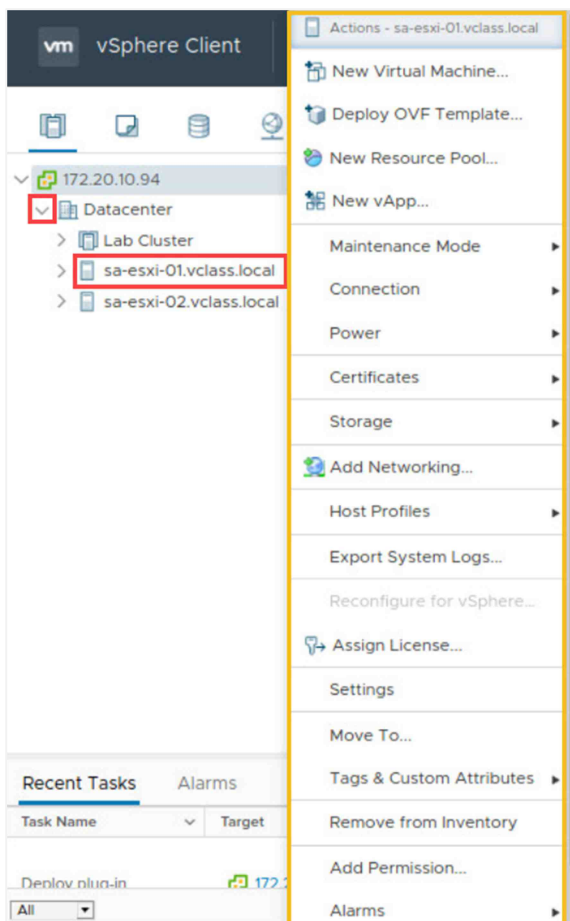
7 Use an Active Directory User to Navigate the vSphere Client and Create a Virtual Machine

In this task, you will create a virtual machine to show how an Active Directory user can perform administrative tasks.

1. In the vSphere Client, from the dropdown Menu, select Hosts and Clusters.



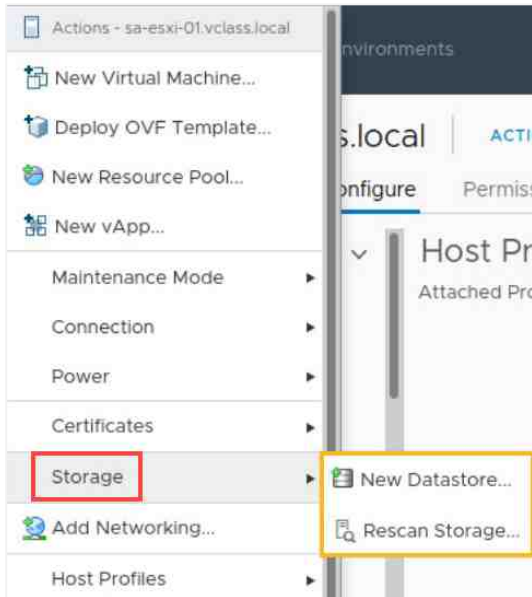
2. Expand the Datacenter object, and right-click sa-esxi-01.vclass.local.



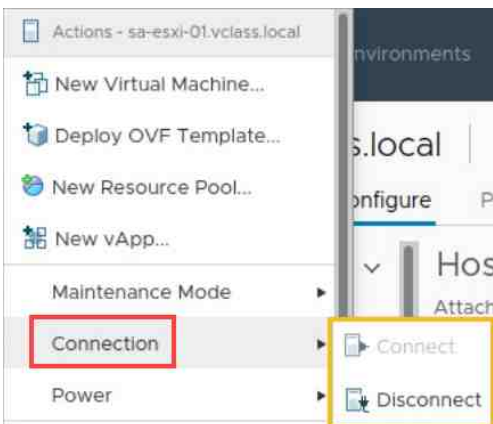
Please Note

Notice that all the Actions for sa-esxi-01 are available. This is because you assigned the sysadmin user with Administrator rights.

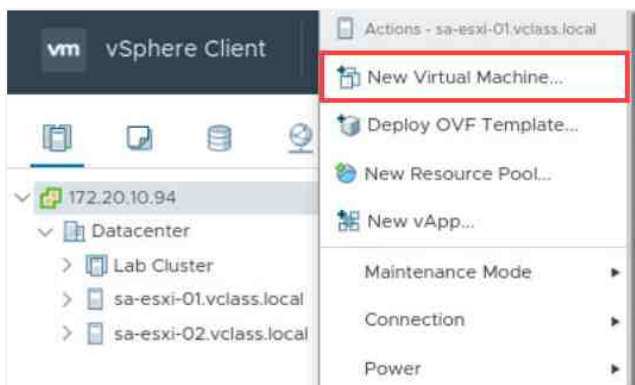
3. In the Actions window for sa-esxi-01, select Storage. Notice that you have permissions to add a New Datastore and Rescan Storage.



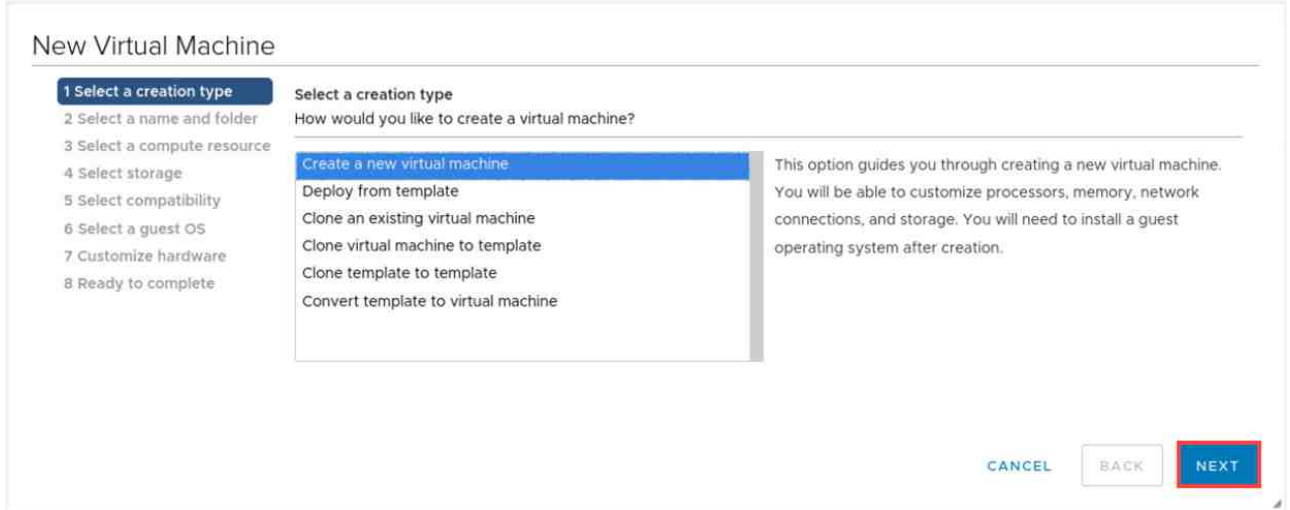
4. In the Actions window, select Connection. Notice that you can Disconnect sa-esxi-01 from vCenter.



5. In the Actions window, click New Virtual Machine.



6. In the New Virtual Machine window, click Next.



The screenshot shows the 'New Virtual Machine' wizard at step 1, 'Select a creation type'. On the left, a list of steps is shown, with step 1 highlighted. The main area has the heading 'Select a creation type' and the question 'How would you like to create a virtual machine?'. A list of options is displayed, with 'Create a new virtual machine' selected and highlighted in blue. To the right of this list, a descriptive text explains that this option guides the user through creating a new VM, allowing customization of processors, memory, network, and storage, and noting that a guest OS must be installed after creation. At the bottom right, there are three buttons: 'CANCEL', 'BACK', and 'NEXT'. The 'NEXT' button is highlighted with a red border.

New Virtual Machine

1 Select a creation type
2 Select a name and folder
3 Select a compute resource
4 Select storage
5 Select compatibility
6 Select a guest OS
7 Customize hardware
8 Ready to complete

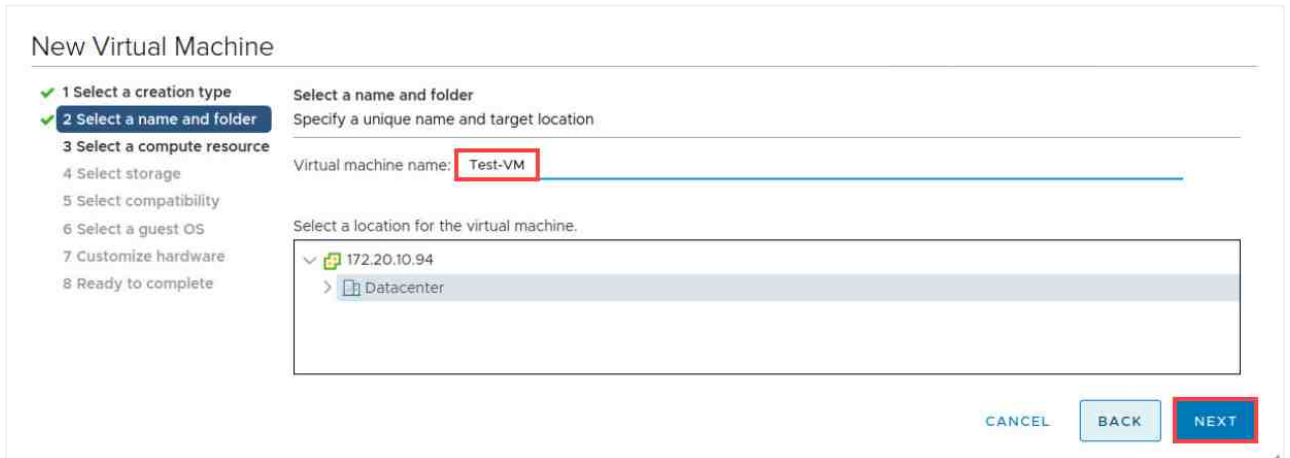
Select a creation type
How would you like to create a virtual machine?

Create a new virtual machine
Deploy from template
Clone an existing virtual machine
Clone virtual machine to template
Clone template to template
Convert template to virtual machine

This option guides you through creating a new virtual machine. You will be able to customize processors, memory, network connections, and storage. You will need to install a guest operating system after creation.

CANCEL BACK NEXT

7. On the Select a name and folder step, enter `Test-VM` in the Virtual machine name text box. Click Next.



The screenshot shows the 'New Virtual Machine' wizard at step 2, 'Select a name and folder'. The left sidebar shows steps 1 and 2 as completed with green checkmarks, and step 3 as the current step. The main area has the heading 'Select a name and folder' and the instruction 'Specify a unique name and target location'. The 'Virtual machine name:' text box contains the text 'Test-VM', which is highlighted with a red border. Below this, there is a section 'Select a location for the virtual machine.' with a tree view showing a folder structure: '172.20.10.94' (expanded) > 'Datacenter'. At the bottom right, there are three buttons: 'CANCEL', 'BACK', and 'NEXT'. The 'NEXT' button is highlighted with a red border.

New Virtual Machine

✓ 1 Select a creation type
✓ 2 Select a name and folder
3 Select a compute resource
4 Select storage
5 Select compatibility
6 Select a guest OS
7 Customize hardware
8 Ready to complete

Select a name and folder
Specify a unique name and target location

Virtual machine name: Test-VM

Select a location for the virtual machine.

172.20.10.94
Datacenter

CANCEL BACK NEXT

8. On the Select a compute resource page, select sa-esxi-01.vclass.local. Click Next.

New Virtual Machine

- ✓ 1 Select a creation type
- ✓ 2 Select a name and folder
- 3 Select a compute resource**
- 4 Select storage
- 5 Select compatibility
- 6 Select a guest OS
- 7 Customize hardware
- 8 Ready to complete

Select a compute resource
Select the destination compute resource for this operation

▼ Datacenter

- > Lab Cluster
 - sa-esxi-01.vclass.local**
 - sa-esxi-02.vclass.local

Compatibility

✓ Compatibility checks succeeded.

CANCEL BACK **NEXT**

9. On the Select a compute resource page, select Local01-2 and click Next.

New Virtual Machine

- ✓ 1 Select a creation type
- ✓ 2 Select a name and folder
- ✓ 3 Select a compute resource
- 4 Select storage**
- 5 Select compatibility
- 6 Select a guest OS
- 7 Customize hardware
- 8 Ready to complete

Select storage
Select the storage for the configuration and disk files

☐ Encrypt this virtual machine (Requires Key Management Server)

VM Storage Policy: Datastore Default ▼

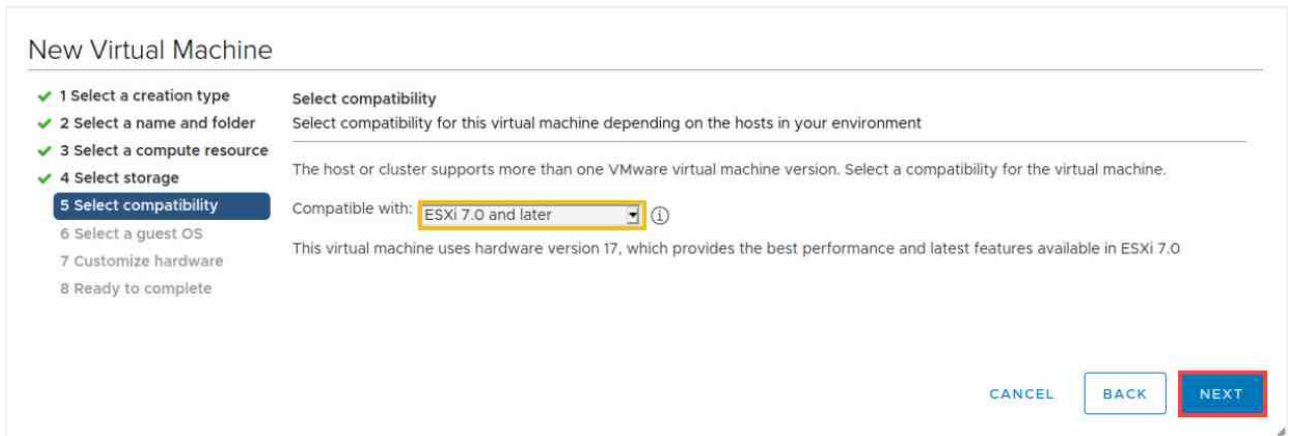
Name	Capacity	Provisioned	Free	Type	Cluster
datastore 1	14.75 GB	12.64 GB	9.79 GB	VMFS 6	
ISCSI-LUN0	35 GB	12.29 GB	22.71 GB	VMFS 5	
Local01-2	14.75 GB	1.41 GB	13.34 GB	VMFS 6	
NFS-1	9.62 GB	0 B	9.62 GB	NFS v3	

Compatibility

✓ Compatibility checks succeeded.

CANCEL BACK **NEXT**

10. On the Select Storage step, leave ESXi-7.0 and later and click Next.



New Virtual Machine

- ✓ 1 Select a creation type
- ✓ 2 Select a name and folder
- ✓ 3 Select a compute resource
- ✓ 4 Select storage
- 5 Select compatibility**
- 6 Select a guest OS
- 7 Customize hardware
- 8 Ready to complete

Select compatibility
Select compatibility for this virtual machine depending on the hosts in your environment

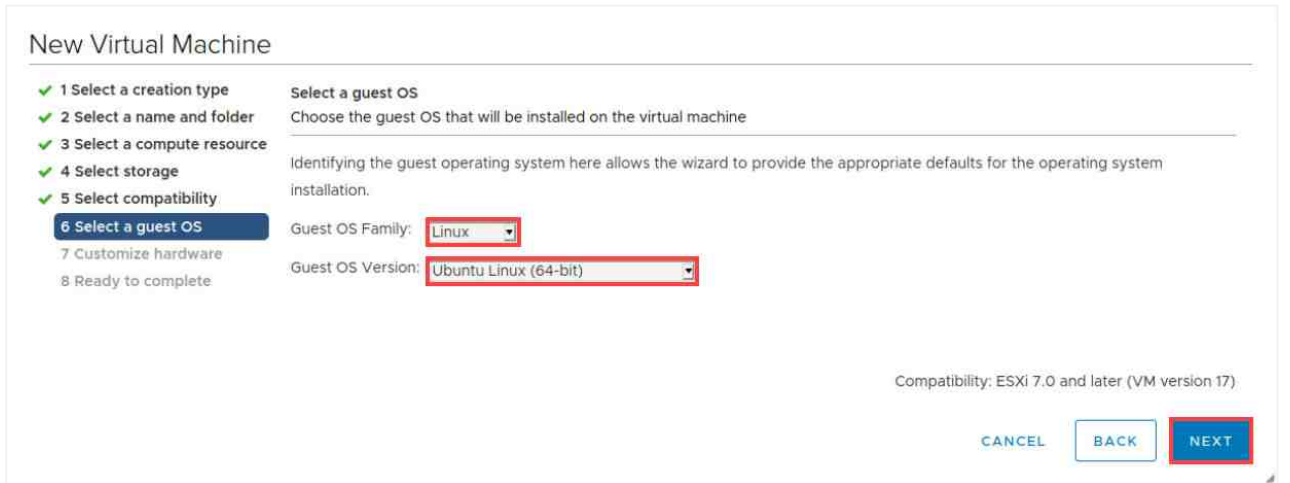
The host or cluster supports more than one VMware virtual machine version. Select a compatibility for the virtual machine.

Compatible with: **ESXi 7.0 and later** ⓘ

This virtual machine uses hardware version 17, which provides the best performance and latest features available in ESXi 7.0

CANCEL BACK **NEXT**

11. On the Select a guest OS step, select Linux from the Guest OS Family dropdown and Ubuntu Linux (64-bit) from the Guest OS Version dropdown. Click Next.



New Virtual Machine

- ✓ 1 Select a creation type
- ✓ 2 Select a name and folder
- ✓ 3 Select a compute resource
- ✓ 4 Select storage
- ✓ 5 Select compatibility
- 6 Select a guest OS**
- 7 Customize hardware
- 8 Ready to complete

Select a guest OS
Choose the guest OS that will be installed on the virtual machine

Identifying the guest operating system here allows the wizard to provide the appropriate defaults for the operating system installation.

Guest OS Family: **Linux**

Guest OS Version: **Ubuntu Linux (64-bit)**

Compatibility: ESXi 7.0 and later (VM version 17)

CANCEL BACK **NEXT**

12. On the Customize hardware step, enter 10 GB for the hard disk size. Click Next.

New Virtual Machine

- ✓ 1 Select a creation type
- ✓ 2 Select a name and folder
- ✓ 3 Select a compute resource
- ✓ 4 Select storage
- ✓ 5 Select compatibility
- ✓ 6 Select a guest OS
- 7 Customize hardware**
- 8 Ready to complete

Customize hardware
Configure the virtual machine hardware

Virtual Hardware VM Options

ADD NEW DEVICE

> CPU	1	
> Memory	1	GB
> New Hard disk *	10	GB
> New SCSI controller *	LSI Logic Parallel	
> New Network *	VM Network	☒ Connect...
> New CD/DVD Drive *	Client Device	☐ Connect...
> Video card *	Specify custom settings	
> Security Devices	Not Configured	
VMCI device		
New SATA Controller	New SATA Controller	

CANCEL BACK **NEXT**

13. On the Ready to complete step, click Finish.

New Virtual Machine

- ✓ 1 Select a creation type
- ✓ 2 Select a name and folder
- ✓ 3 Select a compute resource
- ✓ 4 Select storage
- ✓ 5 Select compatibility
- ✓ 6 Select a guest OS
- ✓ 7 Customize hardware
- 8 Ready to complete**

Datastore	Local01-2
Guest OS name	Ubuntu Linux (64-bit)
Virtualization Based Security	Disabled
CPUs	1
Memory	1 GB
NICs	1
NIC 1 network	VM Network
NIC 1 type	VMXNET 3
SCSI controller 1	LSI Logic Parallel
Create hard disk 1	New virtual disk
Capacity	10 GB
Datastore	Local01-2
Virtual device node	SCSI(0:0)
Mode	Dependent

Compatibility: ESXi 7.0 and later (VM version 17)

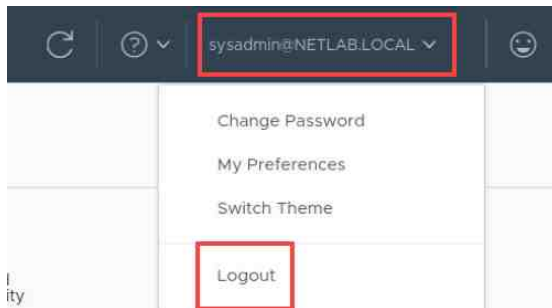
CANCEL BACK **FINISH**

14. Leave the vSphere Client open and continue to the next task.

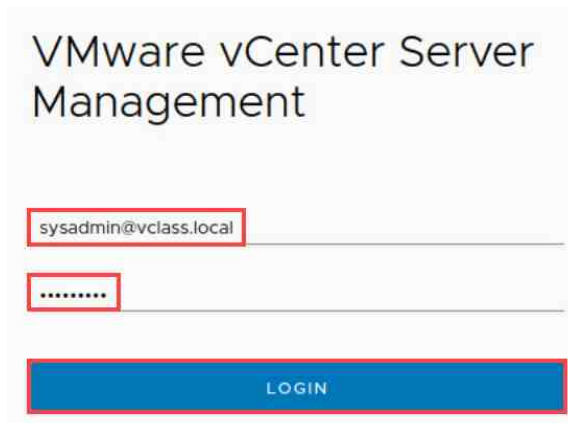
8 Modify Permissions for Active Directory User and Navigate the vSphere Client

In this task, you will modify the permissions for the sysadmin Active Directory user to Read-only. After you change the permissions to Read-only, you will navigate the vSphere Client while noticing that you cannot make any configuration changes.

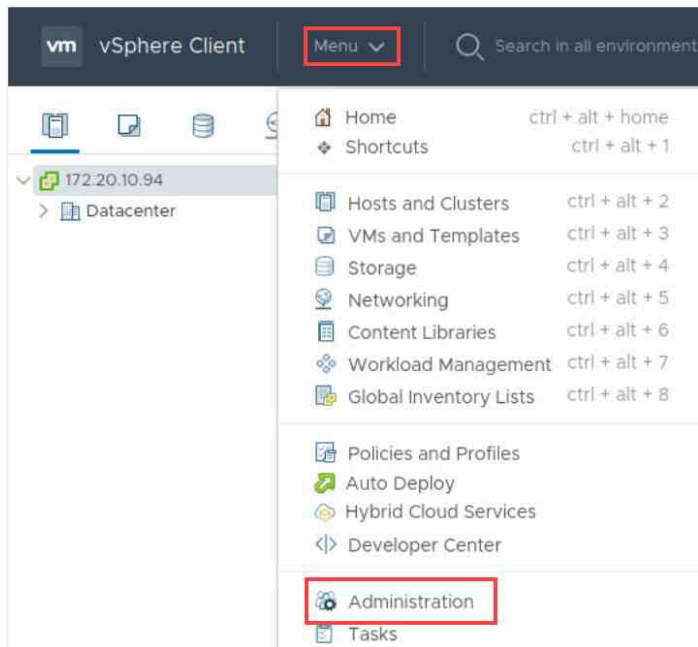
1. Log out of the vSphere Client by selecting sysadmin@NETLAB.LOCAL and click Logout.



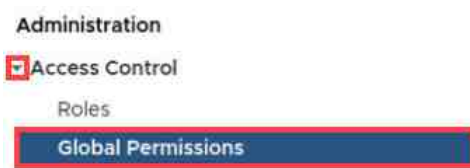
2. On the vSphere Client login screen, enter sysadmin@vclass.local and vmware123 as the password. Click LOGIN.



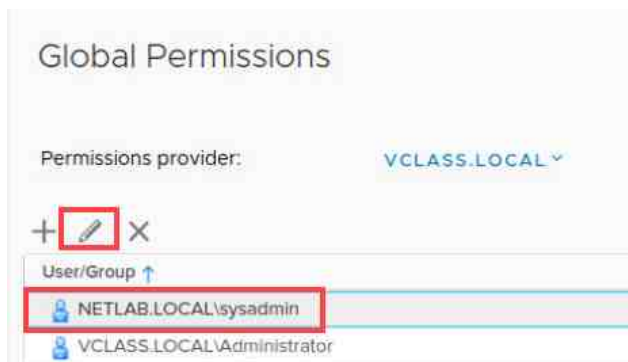
- From the dropdown Menu, select Administration.



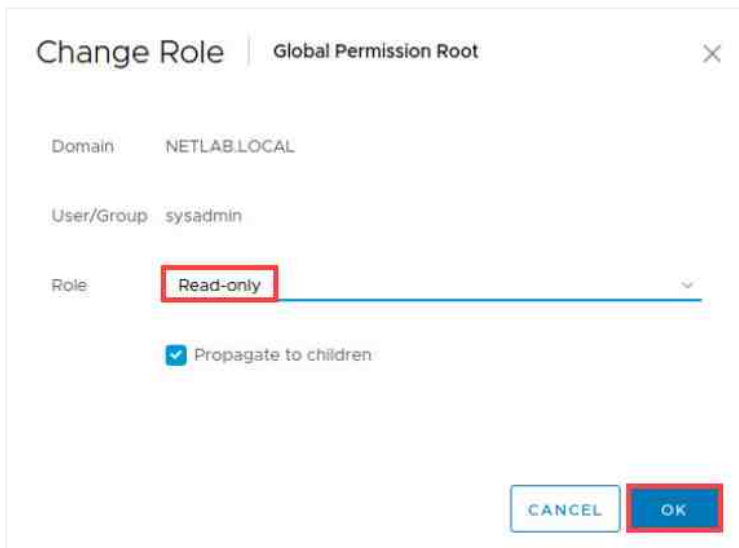
- In the navigation pane, expand Access control and click Global Permissions.



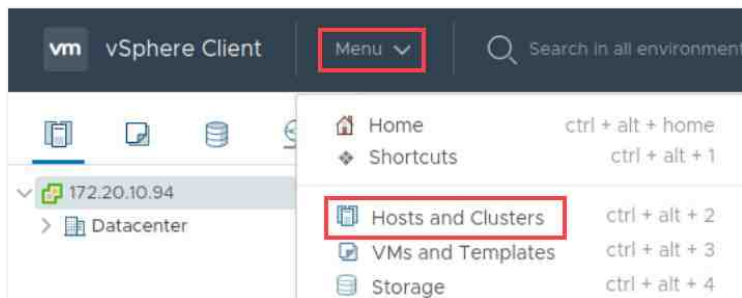
- In the Global Permissions pane, select NETLAB.LOCAL\\sysadmin. Click on the edit icon (pencil).



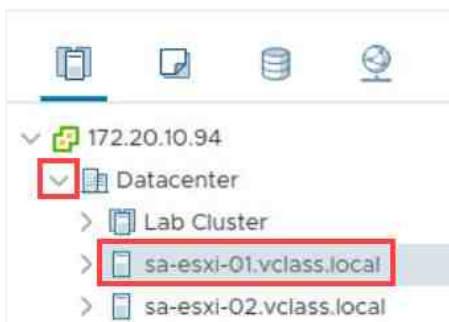
6. In the Change Role window, change the Role to Read-only and click OK.



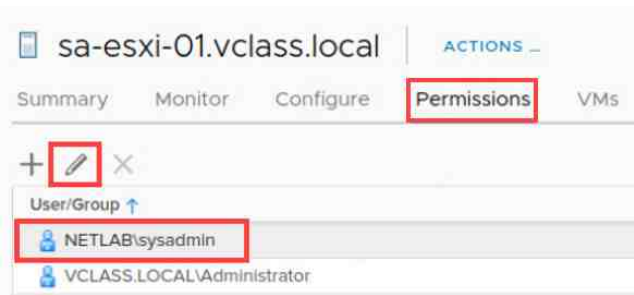
7. From the dropdown Menu, select Hosts and Clusters.



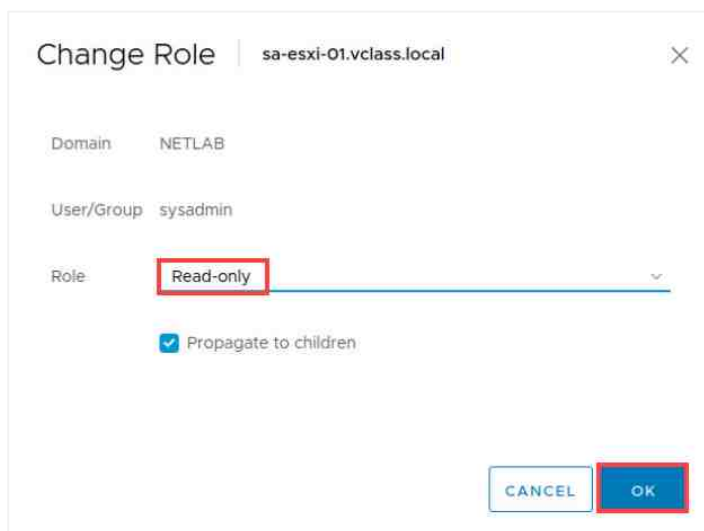
8. In the navigation pane, expand Datacenter and select sa-esxi-01.vclass.local.



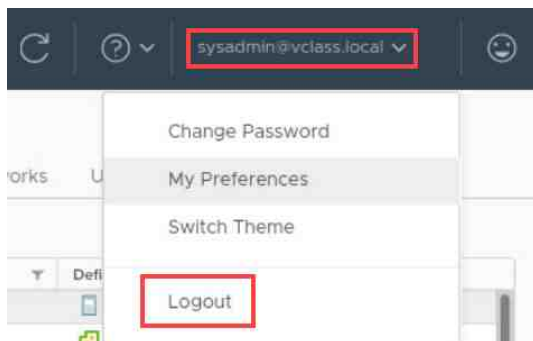
9. In the sa-esxi-01.vclass.local pane, click the Permissions tab. Select NETLAB\sysadmin and click on the edit icon (pencil).



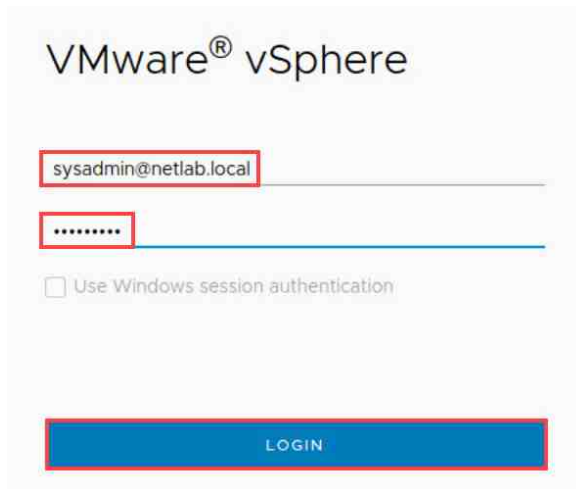
10. In the Change Role window, change the Role to Read-only and click OK.



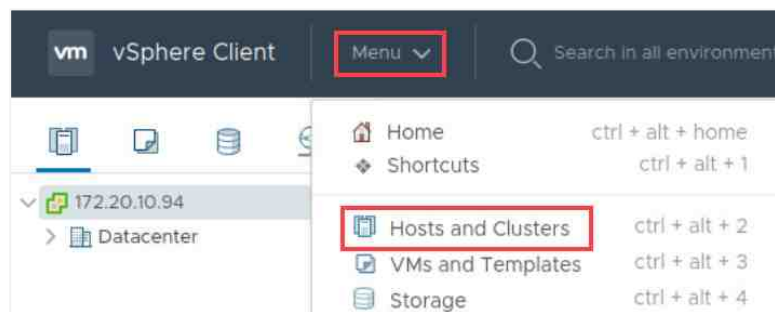
11. Log out of the vSphere Client by selecting sysadmin@vclass.local and click Logout.



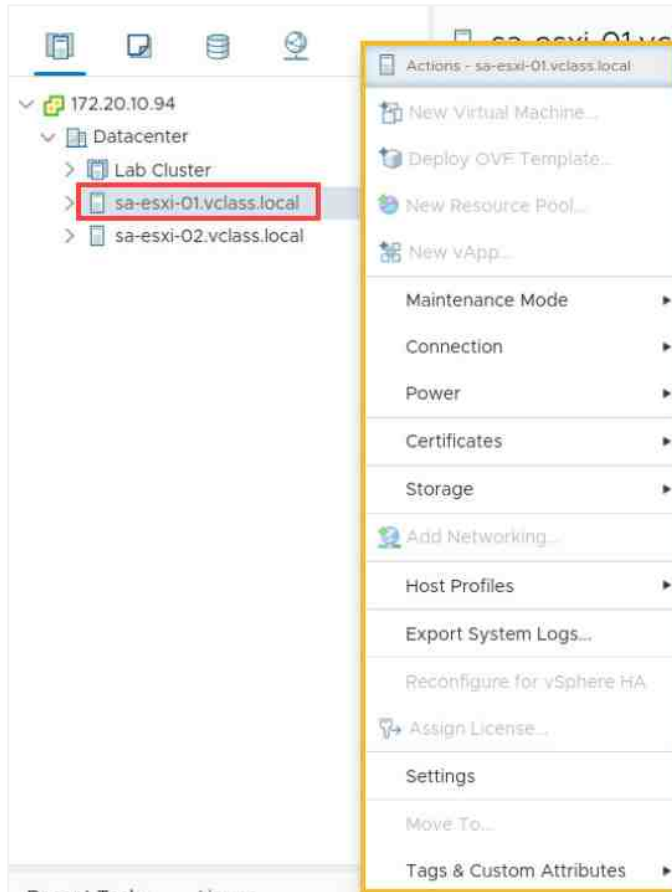
12. On the vSphere Client login screen, enter `sysadmin@netlab.local` and `vmware123` as the password. Click LOGIN.



13. In the vSphere Client, from the dropdown Menu, select Hosts and Clusters.

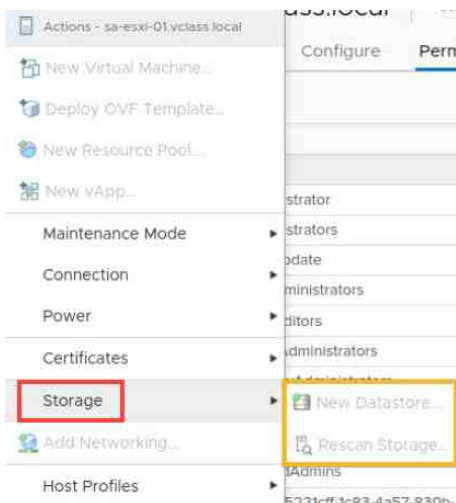


14. Expand the Datacenter object, and right-click sa-esxi-01.vclass.local. Notice there are several configuration options greyed out.

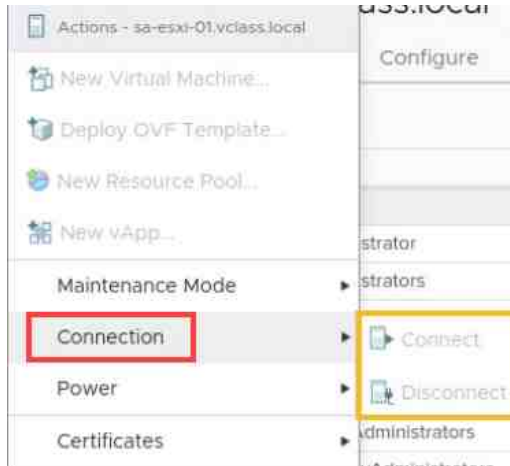
**Please Note**

The configuration options greyed out because you re-assigned the sysadmin user with Read-only rights for sa-esxi-01.vclass.local.

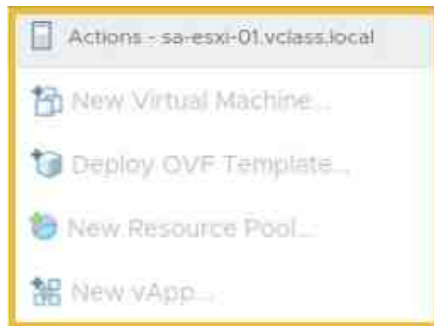
15. In the Actions window for sa-esxi-01, select Storage. Notice the permissions to add a New Datastore and Rescan Storage are now greyed out.



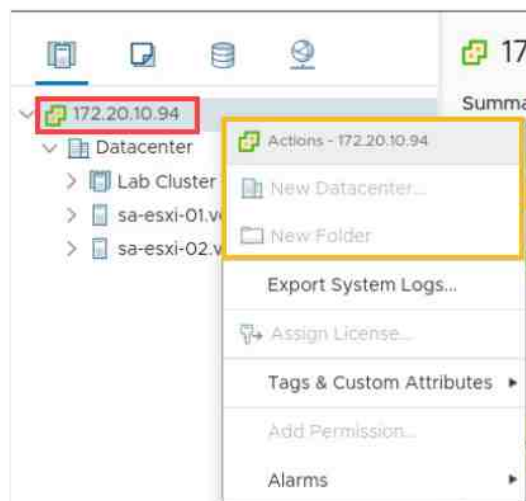
16. In the Actions window, select Connection. Notice the Connection options are now greyed out.



17. In the Actions window, notice that the options to add a New Virtual Machine, Deploy OVF Template, New Resource Pool, and New vApp are greyed out.



18. In the navigation pane, right-click 172.20.10.94. Notice that you cannot add a New Datacenter or a New Folder. This is because you changed the permissions for the vCenter Server Appliance to read-only for the sysadmin user.



19. The lab is now complete; you may end the reservation.